23-202-0601-COMPUTER NETWORKS

Course Outcomes:

On completion of this course the student will be able to:

1. *Familiarize with fundamental underlying principles of computer networking.*
2. *Explain the details and functionality of layered network architecture.*
3. *Apply mathematical foundations to solve computational problems in computer networking.*
4. *Acquire knowledge in ethical, legal, security and social issues related to computer networking.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	2	2							1	2	1	3	1
CO2	3	2	2	2	3							1	2	1	3	1
CO3	3	3	2	3	3							1	3	1	2	2
CO4	2	2	2	2	2							1	1	1	1	2

1-Slightly; 2-Moderately; 3-Substantially

Module I

TCP/IP Protocol stack - Application Layer: Application layer Protocols: - WWW and HTTP, FTP, DNS, SMTP, P2P File sharing, Domain Name System (DNS), Video Streaming and Content Distribution Networks-Internet Video- HTTP Streaming and DASH -Content Distribution Networks , Case Studies: Netflix and YouTube. Socket Programming.

Module II

Transport layer: Transport Layer Services, Relationship with Network Layer, Relationship with Application Layer, Multiplexing and Demultiplexing, UDP, TCP: Header, Segment Structure, Services, Connection establishment and termination, Flow control and window size advertising, TCP timeout and retransmission, Congestion Control, TCP Fairness.

Module III

Network Layer: Network layer Services, Datagram and Virtual circuit services, IP datagram format and Types of Services, The Original Classful Addressing Scheme Dotted Decimal Notation – Subnet and Classless Extensions - Router architectures. IP Multicast Addresses. ARP Protocol. IPv6. Datagram encapsulation and Fragmentation, Reassembly and fragmentation

Module IV

Routing algorithm- LS and DV, Intra-AS Routing in the Internet: OSPF-Routing Among the ISPs: BGP. SDN control plane. ICMP, SNMP.

Data link layer services, Multiple Access Protocols. Switching, VLANs, MPLS, Data centre networking, Software-defined networking.

References:

1. James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach 7th Edition, Pearson Education, ISBN: 978-1292153599.
2. Andrew S. Tanenbaum and David J. Wetherall, Computer Networks, 5th Edition, Pearson Education, 2013, ISBN: 978- 9332518742.
3. Peterson and Davie, "Computer Networks, A Systems Approach", 5th ed., Elsevier,, ISBN:978-0123850591
4. Douglas E. Comer, Computer Networks and Internet, 6th Edition, Pearson Education, ISBN: 978-9352869152.
5. William Stallings, Data and Computer Communication, 10th Edition, Pearson Education, 2006.978-0133506488.
6. Behrouz A. Fourouzan, Firouz Mosharraf, Computer Networks - A Top Down Approach, Tata McGraw Hill, ISBN: 13978-1-25-900156-7.

23-202 -0602 - COMPILER CONSTRUCTION

Course Outcomes:

On completion of this course the student will be able to:

1. *Summarize the functionality of each phase involved in compilation process.*
2. *Develop scanner and parser using lex and yacc tools.*
3. *Design top down parsers including recursive descent parser and non-recursive predictive parser for CFGs.*
4. *Design bottom up parsers including shift reduce, operator precedence and LR parsers (SLR,CLR and LALR).*
5. *Explain Syntax directed translation using S-attributed definition and L-attributed definition.*
6. *Familiarize specification for a type checker and run time environment.*
7. *Comprehend different representations of intermediate code.*
8. *Describe various code optimization techniques to improve the performance of a program and learn code generation techniques.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3													2		
CO2	3		3										3	3	3	
CO3	3	3	3										3	3		
CO4	3	3	3										3	3		
CO5	3															
CO6	3															
CO7	3															
CO8	3															

1-Slightly; 2-Moderately; 3-Substantially

Module I

Compiler: Introduction – Analysis of the source program – phases of a compiler – Lexical analysis –Role of the lexical analyser – Input Buffering -- Specification of tokens – Recognition of tokens –Lexical analyser generators.

Module II

Syntax Analysis – Role of the parser – Context free grammars – Top-down parsing – Bottom-up parsing –Operator precedence parsing – LR parsers (SLR, Canonical LR, LALR) – Parser generators.

Module III

Syntax-directed translation – Syntax-directed definitions – S-attributed definitions –L-attributed definition – Top-down and bottom-up translation – Type checking – Type systems – Specification of a type checker. Run time environment – Source language issues – Storage organization – Storage allocation strategies – Access to nonlocal names – Symbol tables.

Module IV

Intermediate code generation – Intermediate languages – Declarations – Assignment Statement – Boolean Expression – Procedure calls - Code optimization – Introduction – Sources of optimization – Introduction to data flow analysis. Code generator – Issues in the design of a code generator, the target machine, A simple code generator.

References:

1. Alfred V. Aho, Ravi Sethi & Jeffrey D. Ullman, Compilers Principles, Techniques & Tools, Pearson.
2. Kenneth C. Louden, Compiler Construction: Principles and Practice, Thomson Learning, India.
3. Keith D. Cooper & Linda Torczon, Engineering a Compiler, 2nd Edition, Elsevier
4. Muchnick, S.S., Harcourt Asra, Advanced Compiler Design implementation, Morgan Kaufman.
5. Alan Holub, Compiler Design in C, PHI.

23-202 -0603 - ANALYSIS AND DESIGN OF ALGORITHMS

Course Outcomes:

On completion of this course the student will be able to:

1. *Analyse a given algorithm and express its worst, best and average time and space complexities in asymptotic notations and Solve recurrence equations.*
2. *Analyse different searching, sorting algorithms and complex data structures.*
3. *Understand the greedy and dynamic programming paradigm and its algorithmic design solutions.*
4. *Design efficient algorithms using Backtracking and Branch and Bound Techniques for solving problems.*
5. *Classify computational problems into P, NP, NP-Hard and NP-Complete complexity classes.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2								2	3			1
CO2	3	3	2	2								2	3			1
CO3	3	3	3	2	2							2	3			2
CO4	3	3	3	2	2							2	3			2
CO5	2	2	3	3	2							2	1			2

1-Slightly; 2-Moderately; 3-Substantially

Module I

Introduction to algorithm analysis-Time and Space Complexity-Classifying functions by their asymptotic growth rate-Best, Worst and Average case complexities-Complexity Calculation of simple algorithms (sequential and iterative algorithms).Recurrence Equations: Solution of Recurrence Equations –Iteration Method and Recursion Tree Methods, Masters Theorem-divide and conquer and decrease and conquer.

Module II

Analysis of searching Algorithms and Sorting Algorithms- Linear Search and Binary Search. Sorting- Quick Sort, Merge Sort and Heap Sort - Comparison of sorting Algorithms. Analysis of complex Data Structures- Binomial Heap, Fibonacci Heap, Red Black Tree.

Module III

Greedy Strategy- The control Abstraction- Spanning Tree, Minimal Cost Spanning Tree Computation- Prim’s and Kruskal Algorithm-Complexity. Dynamic Programming- The control Abstraction-The Optimality Principle- Optimal matrix multiplication, Bellman-Ford Algorithm.

Back Tracking: -The Control Abstraction – The N Queen's Problem, 0/1 Knapsack Problem. Branch and Bound- Travelling Salesman Problem.

Module IV

Introduction to Complexity Theory- Tractable and Intractable Problems- P and NP, Polynomial Reductions, NP-Hard and NP Complete Complexity Classes. NP Complete Problems- Bin Packing, Graph Colouring, Travelling salesman problem.

References:

1. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, Computer Algorithms, Universities Press, 2007.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, MIT Press, 2009.
3. Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, The Design and Analysis of Computer Algorithms, Pearson Education, 1999.
4. Anany Levitin, Introduction to the Design and Analysis of Algorithms, Pearson, 3rd Edition, 2011.
5. Gilles Brassard, Paul Bratley, Fundamentals of Algorithmics, Pearson Education, 1995.
6. Richard E. Neapolitan, Kumarss Naimipour, Foundations of Algorithms using C++ Psuedocode, Second Edition, 1997.

23-202 -0604 - ARTIFICIAL INTELLIGENCE

Course Outcomes:

On completion of this course the student will be able to:

1. *Explain the characteristics of software agents and java framework for implementing agents.*
2. *Explain production system model and state space search.*
3. *Describe important search techniques and apply to suitable problems.*
4. *Define and apply knowledge representation and deduction methods.*
5. *Explain nonmonotonic reasoning methods*
6. *Describe the important phases in natural language processing.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1		3	3	3	3	3		2					3	3		3
CO2		3	3	3	3	3		2					3	3		3
CO3		3	3	3	3	3		2					3	3		3
CO4		3	3	3	3	3		2					3	3		3
CO5		3	3	3	3	3		2					3	3		3
CO6		3	3	3	3	3		2					3	3		3

1-Slightly; 2-Moderately; 3-Substantially

Module I

Software agents – agent characteristics, agent topology, and agent oriented programming, Java implementation of intelligent agents. AI domains-Problem Characteristics – Problem spaces-search: DFS, BFS - Production systems- Swarm intelligence- genetic algorithm.

Module II

Heuristic search techniques: Generate and Test - Hill climbing -Best first - A* algorithm.
Problem reduction –AO*algorithm, constraint satisfaction - Means Ends analysis. Game playing: Minimax – Alpha-beta cut-off.

Module III

Logic and Deduction: Introduction to symbolic logic - Propositional logic - Well Formed Formula- Predicate Logic - predicates variables and constants - First order logic, Quantifiers-Forward and backward chaining-Resolution by refutation- Unification- Goal trees.

Module IV

Representing Knowledge: Procedural versus Declarative. PROLOG programming Reasoning under uncertainty: Non Monotonic reasoning –support lists and dependency directed backtracking - Statistical reasoning: Bayes theorem. Bayesian networks. Fuzzy Logic, Semantic Nets, Frames, Conceptual Dependency, Scripts, CYC. Natural Language Processing. – machine translation,-Morphology and Finite State Transducers--Word Classes and Part-of-Speech Tagging and chunking-HMM Taggers - Learning: Types of learning- supervised and unsupervised.

References:

1. Elaine Rich and Kevin Knight, Artificial Intelligence, Tata McGraw-Hill, Third Edition, ISBN: 13:978-0-07-008770-5, 2010.
2. Jeffrey M. Bradshaw, Software Agents, AAAI Press/ The MIT Press (1997) (Module 1), ISBN: 0-262-52234-9.
3. Dan W. Patterson, Introduction to Artificial Intelligence and Expert Systems, Prentice Hall India Ltd., 2009, ISBN: 81-203-0777-1 (modules 2,3and 4).
4. Stuart Russell and Peter Norvig, Artificial Intelligence – A Modern Approach. 3rd Edition, Prentice Hall, 2009.
5. Padhy, N.P., Artificial intelligence and intelligent systems, 2010, 0-19-567154-6.
6. Jurafsky D., Martin J.H., Speech and natural language processing, Second Edition, Prentice Hall, 2008, ISBN 10: 0131873210.

23-202 -0605 - CRYPTOGRAPHY AND NETWORK SECURITY

Course Outcomes:

On completion of this course the student will be able to:

1. *Identify various security attacks in the network that are related to the security goals.*
2. *Familiarise the elementary algorithms based on substitution and transportation.*
3. *Familiarise cryptographic algorithms using symmetric and asymmetric keys.*
4. *Familiarise how key exchange (private and public keys) is performed between the sender and receiver.*
5. *Understand how user authentication and message authentication is achieved in a computer network.*
6. *Familiarise with network security protocols used to protect the network against threats.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1															
CO2	2	1	3										2			
CO3	2	1	3										2			
CO4	3	2	3										1			
CO5	2	2	2													
CO6	1	2	2													

1-Slightly; 2-Moderately; 3-Substantially

Module I

Introduction to cryptography, security goals, attacks related to security goals. Mathematics of cryptography: euclidean algorithm, congruence, modular arithmetic, multiplicative inverse. Substitution cipher: caesar, monoalphabetic, playfair, hill cipher, affine ciphers. Transportation cipher, stream and block ciphers.

Module II

Symmetric key encryption: Feistel cipher structure , DES, AES, modes of operations of block cipher, avalanche effect. Public Key Encryption: Primality testing- Miller-Rabin Algorithm, RSA cryptosystem.

Module III

Key Management: symmetric key distribution, KDC, Diffie -Hellman protocol, public key distribution, certificate authority, X.509, PKI. Cryptographic hash functions: message authentication, digital signature, SHA-512, MAC

Module IV

User authentication: one way, mutual authentication, Kerberos. Network security: secure email, PGP, SSL architecture- protocols. IP Security: modes, protocols. Intruders, Malicious programs, worms, viruses. Firewall architectures.

References:

1. Cryptography and Network Security – Principles and Practice: William Stallings, Pearson Education, 6th Edition.
2. Cryptography and Network Security: Forouzan Mukhopadhyay, Mc Graw Hill, 3rd Edition.
3. Cryptography and Network Security: Atul Kahate, Mc Graw Hill, 3rd Edition.
4. Introduction to Network Security: Neal Krawetz, CENGAGE Learning.
5. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.

23-202 -0610 - NETWORKS LABORATORY

Course Outcomes:

On completion of this course the student will be able to:

1. *Familiarise network components and structured cabling.*
2. *Write programs for various communication algorithms.*
3. *Familiarise configuration of various servers and firewalls.*
4. *Do simulations of various network protocols using a network simulator such as ns3.*
5. *Design of communication system using embedded boards.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	1	3							1	1	2	3	1
CO2	3	3	2	2	3							1	3	2	3	2
CO3	3	2	2	2	3							1	2	3	3	2
CO4	3	3	2	3	3							1	3	2	3	3
CO5	3	3	3	3	3							1	3	3	3	3

1-Slightly; 2-Moderately; 3-Substantially

Cycle-I

1. Familiarizing computer network components–
   a) Cables
   b) Connector
   c) Switches and Hub
   d) Router
   e) Network Cards etc.
2. Structured cabling, Creating VLAN using switches and routers, Experiments on subnetting and supernetting.
3. Packet capture and analysis using WireShark.
4. Socket programming–Implement TCP and UDP in UNIX domain, Single chatting program, Multi Chat program using Multithread, Applet chatting.

Cycle- II

1. Program to test error detection and correction codes.
2. Program to test various data compression algorithms.
3. Program to test public key and symmetric key cryptography methods.
4. Program to test various message digest algorithms.

Cycle - lll

1. Simulations of CSMA / CD, Aloha and Slotted Aloha protocols.

2. Simulations to test ARP and RARP.
3. Simulation to test CSMA/CA.
4. Simulations to test congestion and flow control methods in TCP and UDP.
5. Simulations to test various routing protocols.
6. Programs using pcap libraries to packet capture and analysis.
7. Install and configure various servers- file server, ssh server, web server, database server etc.
8. ACL, firewall and use of "iptables".
9. Design of communication system using GSM, 3G,GPS and RFID modules using Raspberrypi, Arduino or Edison Board .

References:

1. Richard Stevens,W., Unix network programming, The Sockets Networking API,Vol.1,3rd edition, Addison-Wesley Professional ISBN:9780131411555.
2. James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach 7 th Edition, Pearson Education, ISBN: 978-1292153599.
3. Douglas E. Comer, Hands-on Networking with Internet Technologies, Pearson Education.
4. Todd Lammle, CCNA: Cisco Certified Network Associate Study Guide,John Wiley and Sons, ISBN:9780470410486.
5. Emad Aboelela, Network Simulation Experiments Manual, The Morgan Kaufmann Series in Networking, Elsevier. ISBN: 9780123852113.
6. Jack L. Burbank, An Introduction to Network Simulator 3(ns3), Wiley-Blackwell. ISBN: 978111815899.

23-202 -0611 - MINI PROJECT

Course Outcomes:

On completion of this course the student will be able to:

1. *Identify project topic of current relevance.*
2. *Explain software development cycle with emphasis on different processes - requirements, design and implementation phases.*
3. *Develop confidence at having conceptualized, designed and implemented a working, medium sized project.*
4. *Learn how to work as a team and to do a working project on time with each student taking responsibility for their part in the project.*
5. *Familiarize document and report preparation.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	2	3	3	3	3				1	2		1	3	3		
CO2	2	1	1		3				1	2		1	3	3		
CO3	2	3	3	3	3				1	2		1	3	3		
CO4	2	1	1		3				1	2		1	3	3		
CO5									1			1		1		

1-Slightly; 2-Moderately; 3-Substantially

The students are expected to develop an application in the field of embedded system / mobile application / any other current relevant topic. They have to do a proper system study and prepare SRS and design documents.

Each batch comprising of 3 to 5 students. Each student shall submit a project report at the end of the semester. The project report should contain the design and engineering documentation and test results. Product has to be demonstrated for its full design specifications. Innovative design concepts, reliability considerations and aesthetics / ergonomic aspects taken care of in the project shall be given due weight.

Guidelines for evaluation:

i) Attendance and Regularity		5
ii) Work knowledge and Involvement		15
iii) End-Semester presentation & Oral examination		10
iv) Level of completion and demonstration of functionality/specifications		10
v) Project Report		10
	Total	50 marks

Note: External projects and R&D projects need not be encouraged at this level. Points (i) & (ii) to be evaluated by the project guide & coordinator and the rest by the final evaluation team comprising of 3 teachers including the project guide.

23-202-0606 TO 23-202-0609 PROFESSIONAL ELECTIVE – II
23-202 -0606(IE) - NEURAL NETWORKS AND DEEP LEARNING

Course Outcomes:

On completion of this course the student will be able to:

1. *Identify the basic concepts of deep learning.*
2. *Analyse the deep learning architectures which are appropriate for various types of learning tasks in different domains.*
3. *Illustrate use of TensorFlow libraries to implement deep neural networks.*
4. *Apply TensorFlow in NLP applications.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2															1
CO2	3	2	2										1			3
CO3	2	2	3	2									1			1
CO4	2	2	3	2									1			2

1-Slightly; 2-Moderately; 3-Substantially

Module I

Introduction to neural networks: Artificial neural networks, Biological neural networks, Basic Models of Artificial Neural Networks: Connections, Learning, Activation functions, Important terminologies of Artificial Neural Networks, McCulloch–Pitts neuron. Perceptron: architecture, algorithm, perceptron learning rule convergence theorem.

Feedforward Networks: Multilayer Perceptron, Gradient Descent, Backpropagation.

Module II

Convolutional Neural Networks: Architectures, convolution / pooling layers.

TensorFlow: Introduction, tensors, tensor properties, basic tensor methods.

CNN in TensorFlow: Applying convolution in TensorFlow, applying pool operations in TensorFlow, Applying Dropout operation in Tensor Flow, Implementation of CNN in TensorFlow.

Module III

Recurrent Neural Networks: Back propagation through time, Bidirectional RNNs, Long Short-Term Memory, Gated Recurrent Units, Bidirectional LSTMs. RNN in TensorFlow: TensorFlow LSTM useful classes and methods, Implementation of RNN in TensorFlow.

Module IV

Applications of Deep Learning to NLP: Introduction, working with bag of words, Implementing TF-IDF, Working with skip-gram embeddings, CBOW embeddings, making predictions with word2vec, using doc2vec for sentiment analysis.

References:

1. Laurene Fausett, Fundamentals of Neural Networks, Prentice Hall, New Jersey, 2007.
2. Ian Goodfellow,Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, 2016.
3. Aurelien Geron, Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media,2019
4. Rodolfo Bonnin, Building Machine Learning Projects with TensorFlow, Packt Publishing, 2016.
5. Nick McClure, Tensorflow Machine Learning Cookbook, Packt Publishing, 2017.
6. Principles of Soft Computing, S N Sivanandam, S N Deepa.

23-202 -0607 - DISTRIBUTED COMPUTING

Course Outcomes

On completion of the course, the student will be able to:

1. Distinguish the theoretical and conceptual foundations of distributed computing.

2. Design client server model and peer to peer model for communication.

3. Recognise issues and challenges in the distributed system design.

4. Identify the problems in developing distributed applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2									3	3			
CO2	3	3	3	2								3		3		
CO3	3	2	3									2		2		
CO4	3	3	2									2				

1-Slightly; 2-Moderately; 3-Substantially

Module-I

Characterization of Distributed systems – Introduction – Challenges , System Models, Interprocess communication, Indirect communication, Distributed Objects and Remote Invocation –Case Study - Java RMI.

Module-II

Operating System Support- The Operating system layer – Protection- Processes and Threads- Operating System architecture. Distributed file Systems- Introduction-File Service architecture– Case study Sun NFS. Name services

Module-III

Time and coordination. Synchronizing physical clocks -logical time and logical clocks. Distributed coordination –distributed mutual exclusion – elections. Replication – basic architectural model – consistency and request ordering.

Module-IV

Distributed DBMS Architecture- Distributed Database Design –Query Decomposition and Data Localization-Distributed transactions – concurrency control in distributed transactions– distributed deadlocks –transaction recovery.

References

1. George Coulouris, Jean Dollimore, Tim Kindberg, Distributed Systems – Concepts and design, Fourth Edition, Pearson Education, 2011. ISBN 978-81-317-1840-7

2. Sunita Mahajan, Seema Shah, Distributed Computing, Oxford University Press, 2010. ISBN: 0-19-806186-2.
3. Andrew S. Tanenbaum, Distributed Operating Systems, Pearson Education, 2011. ISBN: 978-81-7758-179-9.
4. Randy Chow, Theodore Johnson, Distributed Operating Systems and Algorithm Analysis, Pearson Education, 2011. ISBN: 978-81-317-2859-8.
5. TamerOzsu, M., Patrick Valduriez, Principles of Distributed Database Systems, Second Edition, Pearson Education.

23-202 -0608 - DIGITAL IMAGE PROCESSING

Course Outcomes:

On completion of this course the student will be able to:

1. *Outline the basics of image processing.*
2. *Interpret image enhancement techniques.*
3. *Illustrate image restoration and segmentation techniques.*
4. *Infer image compression techniques.*
5. *Identify current technologies and applications of image processing.*
6. *Have a general awareness of DIP use cases in medical domain*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2									3			3			
CO2	2	3	3							2			2	2		
CO3	2	2	2	2						3			3	3		
CO4		3	3	3	3				3			3		2		
CO5				3	3	2	2	2	3		1	3			2	2
CO6					3	2	2	2	3		1	3			2	2

1-Slightly;2-Moderately;3-Substantially

Module I

Introduction-Digital image representation, Fundamental steps in image processing, Components of image processing system. Digital image fundamentals- Sampling and quantization, relationship between pixels. Image transforms-2D DFT, DCT, Walsh-Hadamard transform, Haar Transform, Wavelet Transform.

Module II

Image enhancement-Spatial domain methods-Basic Gray Level Transformations, Histogram Processing. Basics of Spatial Filtering-Smoothing, Sharpening. Frequency Domain Methods-low pass filters, high pass filters, homomorphic filters.

Module III

Image restoration- Degradation model, Inverse filtering, Wiener filter, constrained least squares filtering. Morphological Image Processing- Dilation and Erosion, Opening and Closing. Fundamentals of color image processing- Color models-RGB, CMY, CMYK, and HSI. Pseudocolor Image processing- Intensity slicing, gray level to color transformation.

Module IV

Image Segmentation-Detection of Discontinuities, edge linking and boundary detection, thresholding, region based segmentation, use of motion in segmentation.

Image compression-Error-free Compression-variable length coding, LZW coding, bit plane coding, lossless predictive coding, Lossy compression- lossy predictive coding, transform coding, wavelet coding, Image compression standards. Use cases in medical image processing such as cancer detection, disease diagnosis and so on.

References:

1. Rafael C., Gonzalez & Woods R.E., Digital Image Processing, 3rd edition, Pearson Education, 2008.
2. Anil K Jain, Fundamentals of Digital Image Processing, Prentice Hall India, 2010.
3. William K Pratt, Digital Image Processing, 4th Edition, John Wiley and Sons, 2007.
4. Ioannis Pittas, Digital Image Processing Algorithms and Applications, John Wiley, 2000

23-202 -0609 - INFORMATION RETRIEVAL

Course Outcomes:

1. *Students will get the understanding different Information retrieval model.*
2. *Students will get to know about evaluation methods of the information retrieval model.*
3. *Students will get to know the challenges associated with each topic.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	2	3								1		2		3
CO2	2	3	2	1								1		1	2	
CO3	2			2								1		3	3	

1-Slightly; 2-Moderately; 3-Substantially

Module 1

Introduction to Information retrieval: Information retrieval process, Indexing, Information retrieval model, Boolean retrieval model. Dictionary and Postings: Tokenization, Stop words, Stemming, Inverted index, Skip pointers, Phrase queries

Module 2

Tolerant Retrieval: Wild card queries, Permuterm index, Bigram index, Spelling correction, Edit distance, Jaccard coefficient, Soundex. Term Weighting and Vector Space Model. Evaluation: Precision, Recall, F-measure, E-measure, Normalized recall, Evaluation problems.

Module 3

Latent Semantic Indexing: Eigen vectors, Singular value decomposition, Low rank approximation, Problems with Lexical Semantics. Query Expansion: Relevance feedback, Rocchio algorithm, Probabilistic relevance feedback, Query Expansion and its types, Query drift, Probabilistic Information Retrieval: Probabilistic relevance feedback, Probability ranking principle, Binary Independence Model, Bayesian network for text retrieval

Module 4

XML Indexing and Search: Data vs. Text-centric XML, Text-Centric XML retrieval, Structural terms, Content Based Image Retrieval: Introduction to content Based Image retrieval, Challenges in Image retrieval, Image representation, Indexing and retrieving images, Relevance feedback.

References:

1. Introduction to Information Retrieval by Christopher D. Manning
2. Natural Language Processing And Information Retrieval by Tanveer Siddiqui and U. S. Tiwary